DHBW Ravensburg

Intelligence Engineering

Solving Real World Problems At Scale

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LECTURE II

Introduction to Machine Learning Systems Design

This lecture in one slide

II Introduction to Machine Learning Systems Design

Business and ML Objectives
Requirements for ML Systems
Iterative Process
Framing ML Problems
Mind Versus Data

why this lecture matters, and what you can do after it

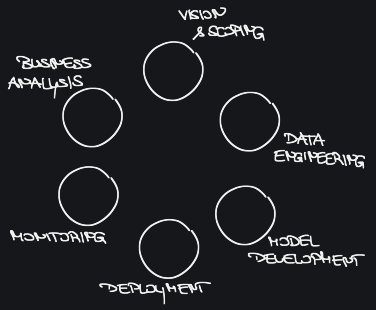
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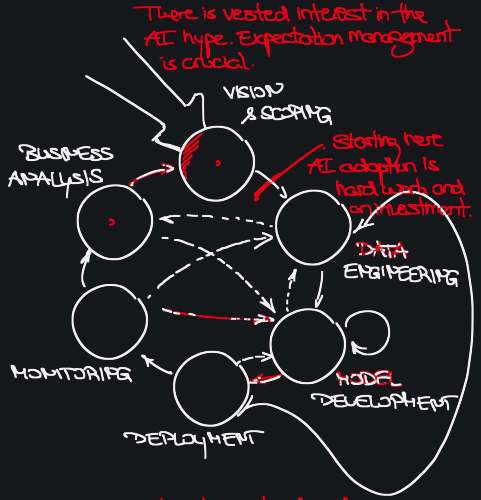
Business and ML Objectives

System Design

ML systems design takes a **system approach** to MLOps: the business requirements, the data stack, infrastructure, deployment, monitoring, and their stakeholders work together to satisfy the specified objectives and requirements.

If this system is built for a business, it must be driven by business objectives, which will need to be translated into ML objectives to guide the development of ML models.

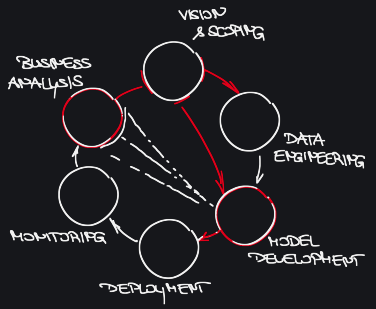



There is vested interest in the AI hype. Expectation management is crucial.
VISION & SCOPING
BUSINESS ANALYSIS
Starting next AI adoption is hard work and an investment.
DATA ENGINEERING
MONITORING
MODEL DEVELOPMENT
DEPLOYMENT
Returns on investment heavily depend on maturity of adoption.
Development cycles and skills don't develop over-time.

Tie ML to business

For an ML project to succeed within a business organization, it's crucial to tie the performance of an ML system to the overall business performance. At best the ML metric becomes a **proxy** for a business metric.

- **Ad click-through rate** every increase results in actual ad revenue.
- **Fraud detection** every fraudulent transaction stopped results in actual money saved.



2

Requirements for ML Systems

- **RELIABILITY**
  Correct function at desired level of performance in the face of adversity.
- **SCALABILITY**
  With respect to complexity, throughput, and other types of growth.
- **MAINTAINABILITY**
  Structure workload, infrastructure and tool landscape so that you can keep up.
- **ADAPTABILITY**
  Capacity to discover potentials for performance improvement and allowing updates without service interruptions.

3

Iterative Process

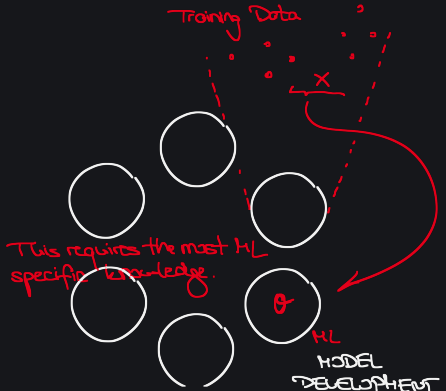
Developing ML systems is a cycle with a lot of back and forth!
#ITERATIVE.
VISION & SCOPING
BUSINESS ANALYSIS
DATA ENGINEERING
MODEL DEVELOPMENT
DEPLOYMENT
MONITORING

Objectives
Key Results
Constraints
VISION
& SCOPING

Stakeholders: users, managers, humanity

We learn from data, so this is where we start with our craftmanship.
Different sources, modalities, formats. for complex
DATA ENGINEERING

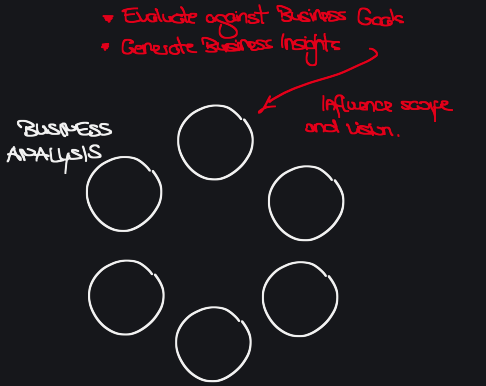
It is all about the labels


Training Data
X
This requires the most ML specific knowledge.
θ
ML
MODEL
DEVELOPMENT

Training Data
X
This requires the most ML specific knowledge.
ML MODEL DEVELOPMENT
• Feature Engineering
• Model Selection
• Training
• Evaluation
ŷ

You will never reach a point where your system is done.
At some point you need to expose your system.
DEPLOYMENT

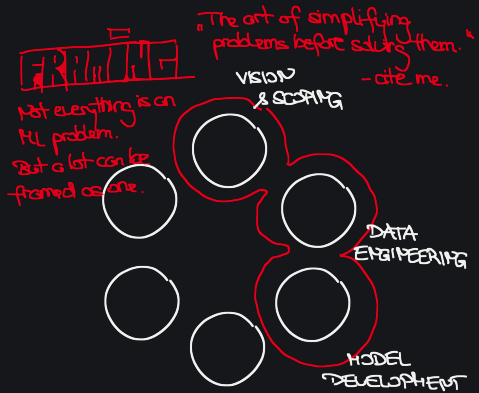
Monitoring: stay adaptive



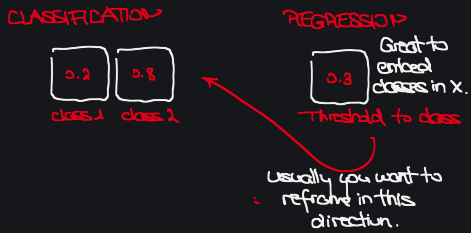
Business analysis closes the loop

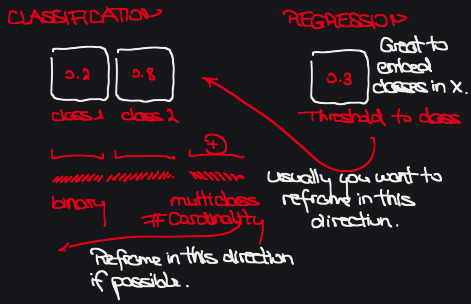
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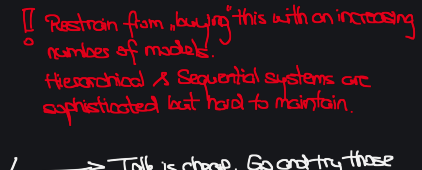
Framing ML Problems



A lot can be framed as an ML problem

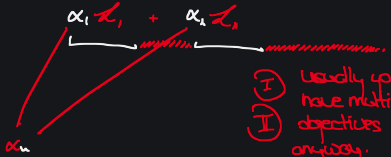

CLASSIFICATION
0.2
0.8
class 1
class 2
REGRESSION
0.3
Great to
extract
classes in X.
Threshold to class
Usually you want to
refrain in this
direction.


CLASSIFICATION
REGRESSION
0.2
0.8
0.3
Great to enforce classes in X.
class 1
class 2
Threshold to class
Usually you want to reframe in this direction.
binary
multiclass
#Cardinality
Reframe in this direction if possible.


Restrain from „buying" this with an increasing number of models.
Hierarchical & Sequential systems are sophisticated but hard to maintain.
Talk is cheap. Go and try those things on MNIST to sharpen your own intuition.

The objective function defines what optimal is

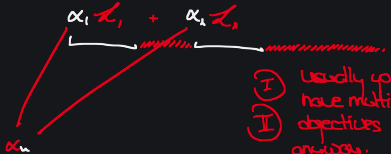
THE EFFECTIVENESS OF DECOUPLING OBJECTIVES.



Usually you have multiple objectives anyway.

This is a clean way to state them explicitly.

THE EFFECTIVENESS OF DECOUPLING OBJECTIVES.



I) Usually you have multiple objectives anyway.

II) This is a clean way to state them explicitly.

#Pareto Optimization

Trivial to weight objectives while training.

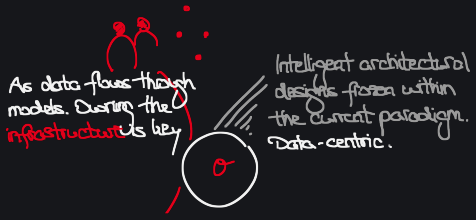
Everytime you touch is a retrain.

Beautiful opportunity to train two models and adaptively weight during inference.

- Objective Functions require gradients.
- For everything else we still need metrics.

5

Mind Versus Data


As data flows through models. During the infrastructure is key
Intelligent architectural designs flourish within the current paradigm.. Data-centric.

Data: model success depends on it

"We don't have better algorithms. We just have better data." - Norvig (Director Search @Google)



Remember how more data is a regularization method, driving generalization of your models?

6

Summary

Summary

References

1 Chip Huyen. *Designing Machine Learning Systems: An Iterative Process for Production-Ready Applications*. Sebastopol, CA: O'Reilly Media, 2022.